

Age Classification using a Regularized ResNet-18 with Knowledge Distillation

A **ResNet-18 architecture** was used as the primary model, strictly following the assignment constraint of training the network **from scratch (weights=None)**. The backbone architecture was kept unchanged, while the classification head was slightly modified by adding a **Dropout layer followed by a linear layer** to improve generalization and reduce overfitting.

Data Augmentation and Regularization

To improve robustness under limited data conditions, several augmentation techniques were applied during training. These included **random resized cropping, horizontal flipping, and mild color jittering**. Additionally, **Random Erasing** was used to randomly mask small image regions so that the model learns broader facial representations rather than relying on specific localized features.

Further regularization was introduced through **MixUp augmentation**, where pairs of training images are linearly combined to create new samples with interpolated labels. This approach encourages smoother decision boundaries and improves the model's ability to generalize. **Label smoothing** was also applied within the cross-entropy loss to prevent overly confident predictions.

Knowledge Distillation

To enhance feature learning while respecting the assignment constraints, **knowledge distillation** was incorporated. A pretrained **CLIP foundation model** was used to generate soft probability distributions for each training image using prompts describing “young” and “old” faces. These soft predictions served as auxiliary supervision.

The student model (ResNet-18) was trained using a combination of **cross-entropy loss** and **Kullback–Leibler divergence loss**, allowing the network to learn both from the ground-truth labels and the semantic guidance provided by CLIP.

Training Strategy

Training was performed using the **Adam optimizer with weight decay**. A **warm-up cosine learning rate schedule** was employed to stabilize early training and gradually decrease the learning rate for improved convergence. Validation accuracy was monitored during training and the best performing model was saved.

As required by the assignment instructions, a **final training run was performed using both the training and validation datasets combined** before generating the submission model.

The final trained network was saved as **b23cm1003.pth**, which can be directly loaded and evaluated using the provided evaluation script.